

AIR UNIVERSITY ISLAMABAD CAMPUS

Visual Programming Lab (CS-284L)

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ThreatCast PK

Live Crowdsourced Cyberattack Intelligence Map of Pakistan

PROJECT PROPOSAL

CLO-2 | GA-5 | Marks: 50

Field	Details
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1. Introduction

THE REAL PROBLEM

Pakistan's digital infrastructure is under relentless and growing attack. In a single year, Kaspersky recorded over 500,000 cyberattacks against Pakistani targets. NCCIA officials confirmed a 35% surge in cybercrime in 2025 alone. The PTA Cybersecurity Report 2024-25 documented a 17% rise in attacks against critical national systems. Yet despite this escalating crisis, there is currently zero publicly accessible, real-time threat intelligence platform for Pakistan.

When an attack hits a Pakistani bank, university, hospital, or telecom operator today, the response is fragmented. CERT-PK issues text advisories buried in government portals that few people read. Small businesses, educational institutions, and individual users have no situational awareness whatsoever. Security teams cannot see emerging patterns. Journalists cannot report on trends. Students and researchers cannot study the threat landscape. The country is effectively flying blind.

ThreatCast PK directly addresses this gap. It is a live, visual, crowdsourced cyberattack intelligence platform that aggregates threat data from open APIs, community reports, and official advisories, and presents it on a real-time animated map of Pakistan. For the first time, any Pakistani — from a SOC analyst at a major bank to a student at a provincial university — can open a browser and see what is being attacked, where, how severely, and what the trend looks like over time.

STATISTICS MOTIVATING THE PROJECT

Metric	Data Point
Cyberattacks/year	500,000+ attacks recorded against Pakistan in a single year (Kaspersky 2023)
2025 Crime Surge	+35% surge in cybercrime in Pakistan in 2025 alone (NCCIA)
Critical Systems	+17% rise in attacks on critical infrastructure (PTA Report 2024-25)
Public Awareness	Zero publicly accessible real-time threat intelligence maps for Pakistan currently exist

PROJECT OBJECTIVE

ThreatCast PK is built in Blazor Server using SignalR for real-time push updates. It integrates live external threat intelligence APIs (AbuseIPDB, GreyNoise), machine learning for anomaly detection and campaign clustering, and a community crowdsourcing layer for locally-reported incidents. The platform serves as both a practical security tool and a definitive demonstration of Blazor Server's capabilities for real-time, data-intensive web applications.

2. Features and CRUD Operations

CORE FEATURES OVERVIEW

ThreatCast PK is organized around four functional pillars: (1) Real-Time Threat Visualization, (2) Crowdsourced Community Reporting, (3) Threat Intelligence Feed Integration, and (4) Machine Learning Analytics.

2.1 Real-Time Threat Map (Blazor Server + SignalR)

The centerpiece of ThreatCast PK is an animated, interactive map of Pakistan rendered via Leaflet.js through Blazor's JavaScript interop. Attack events appear as animated markers on the map in real time as they are reported or detected. Each marker is color-coded by attack type and sized proportionally to severity.

- Attack markers color-coded by type: Ransomware (red), Phishing (orange), DDoS (yellow), Identity Theft (purple), Malware (pink)
- Sector-based filtering: Banking, Telecom, Healthcare, Education, Government, Energy
- City-level geolocation for all major Pakistani cities (Karachi, Lahore, Islamabad, Peshawar, Quetta, Faisalabad, Multan, Rawalpindi)
- Animated pulse effect on active attack markers; fading effect as events age
- Click-to-inspect: clicking any marker opens a detail panel with full event data
- Time-window filter: Last 1 hour / 6 hours / 24 hours / 7 days / 30 days

2.2 Crowdsourced Attack Reporting (CRUD Operations 1-3)

Any verified user can submit an anonymized attack report. Reports are moderated before appearing on the public map.

CRUD 1 — Create: Submit Attack Report

Users fill out a multi-field Blazor form to report a cybersecurity incident they have experienced or observed. Fields include: Attack Type (dropdown), Target Sector (dropdown), City (dropdown), Description (textarea), Impact Level (1-5 scale), Date/Time of incident, and optional contact email for follow-up. On submission, the report enters a pending state awaiting admin moderation. A CAPTCHA challenge prevents automated spam.

CRUD 2 — Read: Browse Submitted Reports

Authenticated moderators can view the full table of submitted reports in the admin panel, filterable by status (Pending / Approved / Rejected), city, attack type, and date range. Public users can browse approved reports in a feed view below the map.

CRUD 3 — Update/Delete: Moderation Panel

Admin users can approve, reject, or edit submitted reports. Approved reports are pushed to the live map via SignalR. Rejected reports are soft-deleted with a reason logged for audit. Moderators can also edit the attack type classification if the user submitted incorrectly.

2.3 Registered User Management (CRUD Operations 4-5)

ThreatCast PK has a full user account system with role-based access control: Public (read-only), Reporter (can submit events), Analyst (can annotate events), and Admin (full access).

CRUD 4 — Create/Read: User Registration and Profile

New users register with email, username, and password. Passwords are hashed with BCrypt. Registered users gain the Reporter role and can submit attack events. User profiles display their submission history, reputation score, and badges earned for consistent accurate reporting.

CRUD 5 — Update/Delete: Profile Management and Admin User Control

Users can update their own profile data (display name, email, notification preferences). Admins can promote users to Analyst or Admin roles, suspend accounts for abuse, or permanently delete accounts. All account actions are logged in an audit trail.

2.4 Threat Intelligence Feed Integration (AbuseIPDB API)

ThreatCast PK integrates with AbuseIPDB's free community API to pull data on malicious IP addresses and enrich crowdsourced reports with external intelligence. GreyNoise API (free community tier) is used to distinguish internet background noise from targeted attacks.

- Automatic IP reputation lookup when a report includes a source IP
- AbuseIPDB confidence score displayed on event detail cards
- CERT-PK advisory feed parsed from public RSS and displayed as official event overlays on the map
- Daily automated pull of the top 50 reported malicious IPs affecting Pakistan, stored in the database for trend analysis

2.5 Threat Alerts and Notification System (CRUD Operations 6-7)

CRUD 6 — Create/Read: Custom Alert Subscriptions

Authenticated users can create custom alert subscriptions: define a combination of attack type, sector, city, and minimum severity level. The system monitors incoming events and sends email notifications when events matching the subscription criteria are detected. Users can view all their active subscriptions in a dashboard panel.

CRUD 7 — Update/Delete: Subscription Management

Users can modify existing alert subscriptions (change thresholds, add/remove cities) or delete subscriptions they no longer need. Admins can create system-wide broadcast alerts that notify all registered users — used for major incidents like confirmed national-level DDoS campaigns.

2.6 Machine Learning Analytics Layer

A Python FastAPI microservice provides ML-powered analytics that are surfaced in the Blazor UI via API calls.

- Isolation Forest campaign detection: if 15+ reports share the same IP range within 6 hours, the system automatically flags a coordinated campaign with a red HIGH ALERT banner on the map
- Sector Risk Score: each sector receives a live risk rating (LOW / MEDIUM / HIGH / CRITICAL) updated in real time as events come in — example: Banking sector: HIGH RISK (42 events in last 24h)
- Time-series trend chart: 30-day rolling attack volume by sector, powered by regression trend analysis
- Attack heatmap by city: normalized density visualization across Pakistani cities

2.7 Threat Pulse Live Feed

A real-time scrolling ticker at the bottom of the dashboard displays incoming threat events as they are pushed to the system — functioning like a cybersecurity Bloomberg terminal. Each item shows: timestamp, city, attack type, sector, and severity. The feed is powered by SignalR and updates without any page refresh.

2.8 Historical Analytics Dashboard

A dedicated analytics page provides historical data visualizations using Chart.js via Blazor JavaScript interop.

- Bar chart: attack volume by week for the last 12 weeks
- Pie chart: attack distribution by type (current month)
- Line chart: sector risk score trends over 30 days
- Table: top 10 most targeted cities with attack counts and dominant attack types
- Export to CSV for all chart data

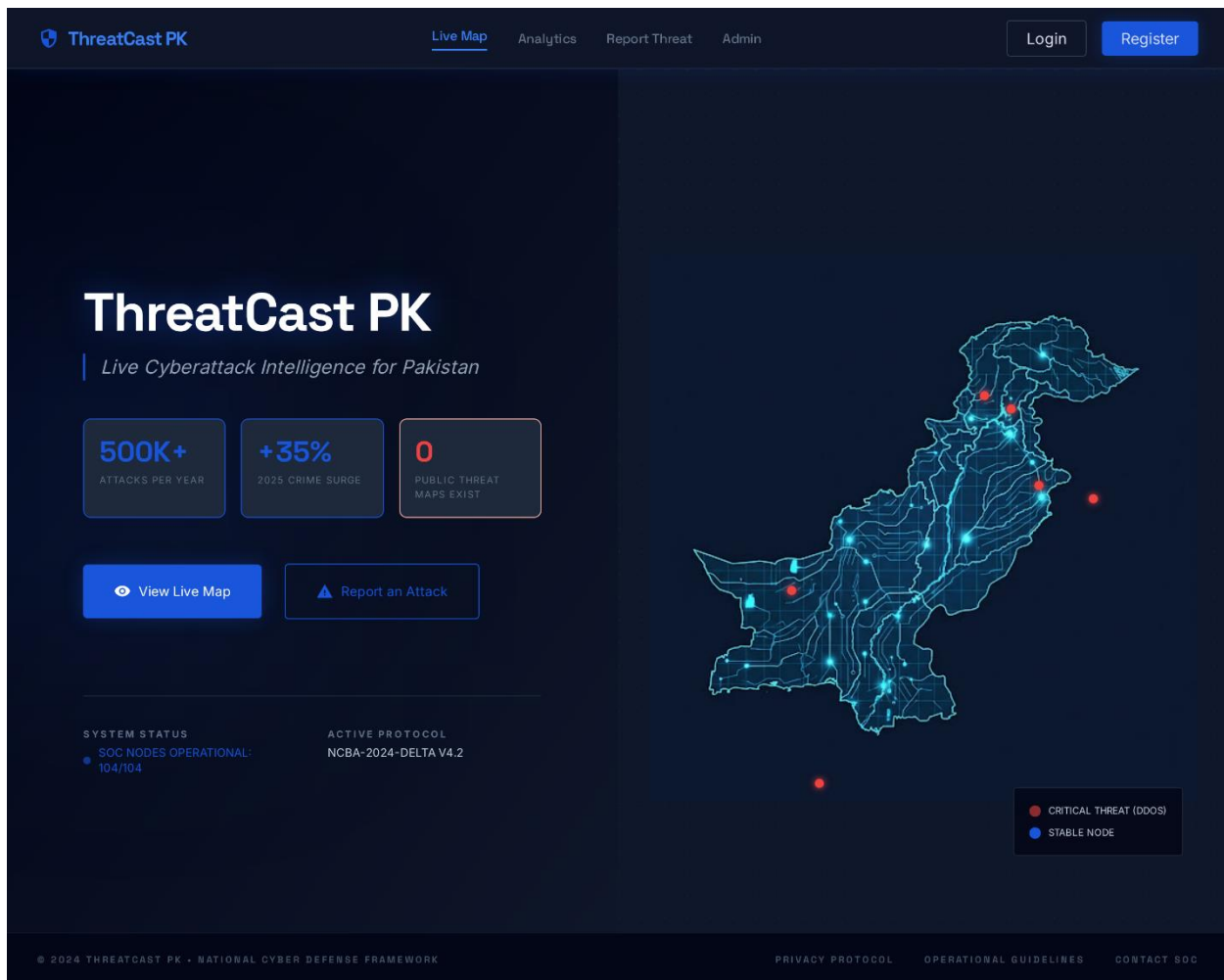
2.9 CRUD Operations Summary Table

#	Operation	Entity	Description	Who
1	CREATE	Attack Report	Submit a new crowdsourced attack event	Registered User
2	READ	Attack Report	Browse & filter all submitted reports	Moderator / Public
3	UPDATE / DELETE	Attack Report	Approve, reject, or edit pending reports	Admin / Moderator
4	CREATE / READ	User Account	Register account, view profile & submission history	Any User
5	UPDATE / DELETE	User Account	Edit profile; admin manages roles & suspensions	User / Admin
6	CREATE / READ	Alert Subscription	Create custom alert rules; view active subscriptions	Registered User
7	UPDATE / DELETE	Alert Subscription	Modify alert thresholds; delete subscriptions	User / Admin

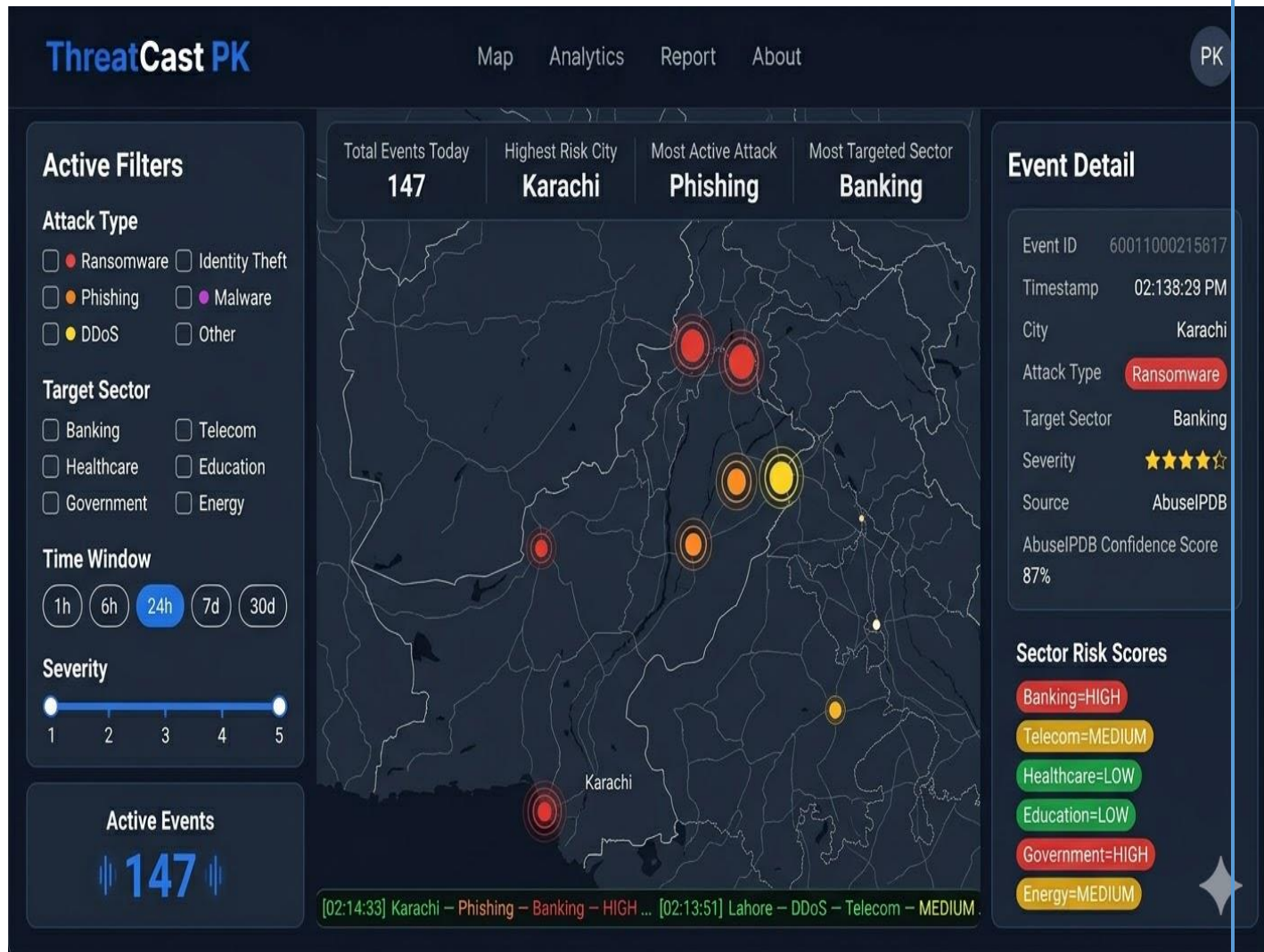
3. Mockup Screen Descriptions

The following descriptions are written for Google Stitch (or Figma) implementation. Each screen description covers layout, components, color usage, and interactive elements. Copy each description block directly into Google Stitch as your prompt.

Screen 1: Landing / Login Page



Screen 2: Main Dashboard / Live Threat Map



Screen 3: Report an Attack (Submit Form)

Report a Cybersecurity Incident

Your report helps protect Pakistan. All submissions are anonymized.

SECTION 1 Incident Details

Attack Type

Ransomware

Target Sector

Government

City

Islamabad

When did this occur?

mm/dd/yyyy, --:-- --

SECTION 2 Impact

Impact Level

1 2 3 4 5

Minor Moderate Critical

Describe the Incident (optional)

Provide technical details or sequence of events...

0 / 500

SECTION 3 Source (Optional)

Source IP Address (if known) ⓘ

e.g. 192.168.1.1

SECTION 4 Contact (Optional)

Your email (for follow-up only — never shared)

name@organization.pk

SECTION 5 Verification

☐ I'm not a robot

RECAPTCHA

Privacy Terms

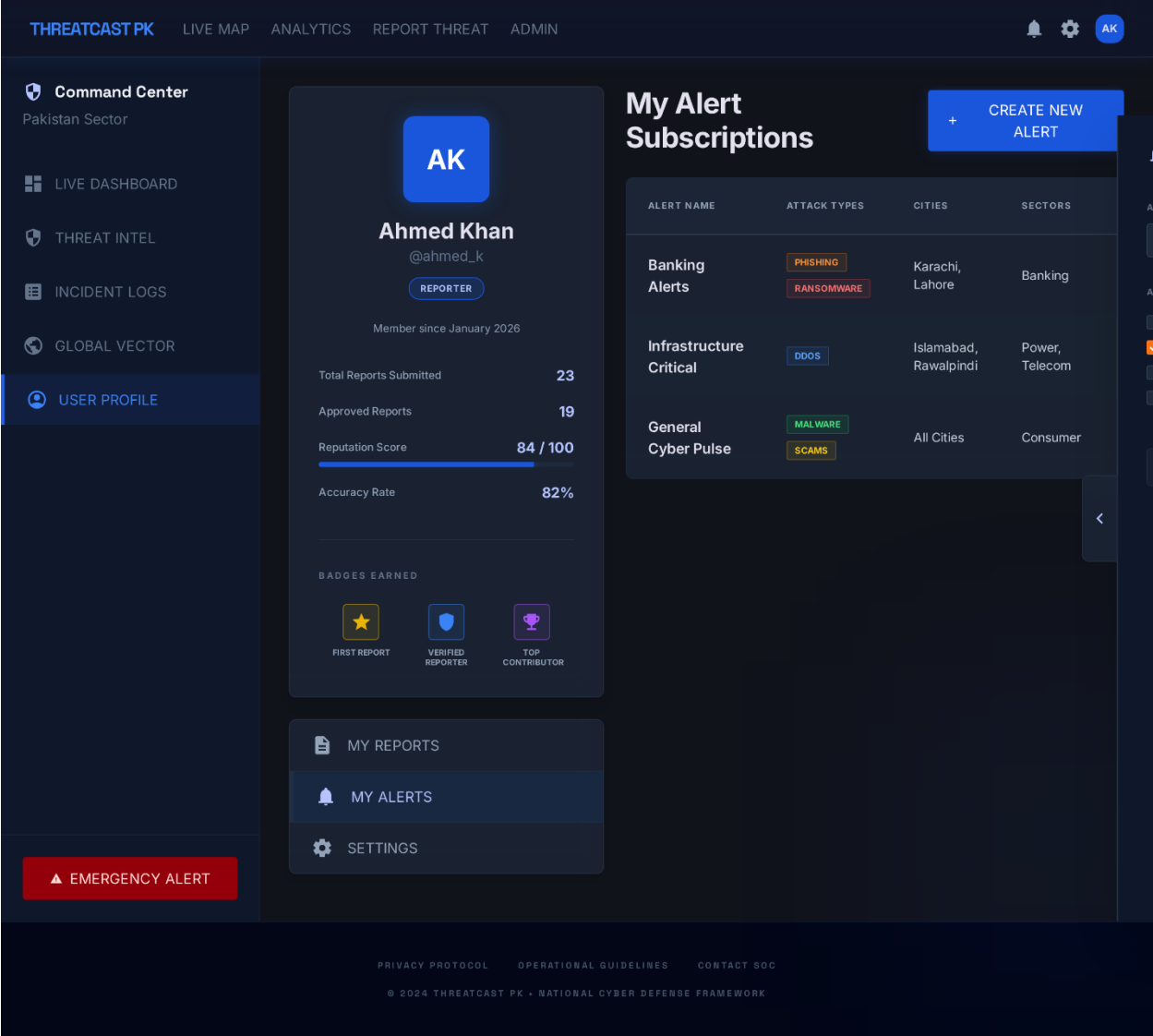
Submit Report

Reports are reviewed by moderators before appearing on the map.
Thank you for contributing to Pakistan's cyber defense.

Screen 4: Analytics Dashboard



Screen 5: User Profile and Alert Subscriptions



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Screen 6: Admin Moderation Panel

ADMIN PANEL – Moderation Queue

[Map](#)
[Analytics](#)
[Report](#)
[About](#)

PK

Pending Review: 14

Approved Today: 67

Rejected Today: 8

Total Reports: 3,847

🔍

☒ 3 selected

Bulk Approve

Bulk Reject

Export Selected

☐	Report ID	Submitted At	City	Attack Type	Sector	Severity	Description	Submitter	Source	Status	Actions
☐	TK-1040	2026-04-27 10:15:30	Karachi	Phishing	Banking	★★★	Suspicious email to staff ...	User#2109	Reported by User	Pending	✓ ✕ ✎
☒	TK-1041	2026-04-27 10:15:30	Lahore	DDoS	Telecom	★★★★★	Network outage affecting ...	User#2110	Reported by User	Pending	✓ ✕ ✎
☒	TK-1042	2026-04-27 10:15:30	Bahawalpur	Malware	Healthcare	★★★★★	Multiple systems locked ~80 chars ...	User#2111	Reported by User	Pending	✓ ✕ ✎
☒	TK-1043	2026-04-27 10:15:30	Islamabad	Ransomware	Education	★★★★★	Server encryption with. ~80 chars ...	User#2112	Reported by User	Approved	✎ ⌛ 🗑
☐	TK-1044	2026-04-27 10:15:30	Multan	Other	Energy	★★	Minor port scanning ...	User#2113	Reported by User	Approved	✎ ⌛ 🗑
☐	TK-1045	2026-04-27 10:15:30	Rawalpindi	DDoS	Government	★	Brief service hiccup ~80 chars ...	User#2114	Reported by User	Rejected	✎ ⌛ ⚡

4. Technology Stack

Layer	Technology	Purpose
Frontend	Blazor Server (.NET 8)	UI components, real-time rendering, form handling
Real-Time	SignalR	Push attack events to all connected clients
Map	Leaflet.js (JS Interop)	Interactive Pakistan map with animated markers
Charts	Chart.js (JS Interop)	Line, bar, pie, and heatmap visualizations
Backend API	ASP.NET Core Web API	CRUD endpoints, authentication, API gateway

ML Microservice	Python FastAPI + scikit-learn	Isolation Forest campaign detection, risk scoring
Database	PostgreSQL + TimescaleDB	Time-series attack event storage and querying
Auth	ASP.NET Core Identity + JWT	User registration, login, role-based access
External API 1	AbuseIPDB API (Free)	IP reputation lookups for reported source IPs
External API 2	GreyNoise Community API	Internet background noise vs. targeted attack classification
Hosting	Render.com / Railway (Free)	Zero-cost deployment for all tiers
DB Hosting	Supabase (Free Tier)	Managed PostgreSQL at PKR 0 cost

5. Iteration / Sprint Plan

The project is structured across 5 two-week sprints running from late April to late June 2026. Each sprint has clearly defined deliverables and acceptance criteria.

Sprint	Dates	Deliverables	Owner	Risk
1	Apr 28 - May 11	Project scaffolding (Blazor Server + ASP.NET Core). Database schema design and Entity Framework models. User auth (Registration, Login, Roles). Static Pakistan map with Leaflet.js. Basic CRUD: Report submission form (Create) and Report list view (Read). Seed database with 50 historical attack events from public CERT-PK reports.	Both	LOW
2	May 12 - May 25	SignalR hub integration — live map updates on new report approval. Admin Moderation Panel (CRUD 3: Update/Delete reports). AbuseIPDB API integration — IP lookup on report submission. Map marker animations and color-coding by attack type. Threat Pulse live ticker component. Alert Subscription Create/Read (CRUD 6).	Both	MEDIUM
3	May 26 - Jun 8	Python ML microservice: Isolation Forest campaign detection. Sector Risk Score calculation and API endpoint. GreyNoise API integration. Alert Subscription Update/Delete (CRUD 7). User profile management page (CRUD 5). Email notification system for alert triggers.	Both	MEDIUM
4	Jun 9 - Jun 22	Analytics Dashboard: time-series charts, sector trends, city heatmap. Chart.js integration via Blazor JS interop. ML campaign alert cards on analytics page. CSV export for event log data. Historical data queries optimized with TimescaleDB. All CRUD operations end-to-end tested.	Both	LOW
5	Jun 23 - Jun 30	Full deployment to Render.com / Railway + Supabase. UI/UX polish: responsive design, loading states, error handling, empty states. Demo preparation: scripted walkthrough of all 7 CRUD operations. Demo data seeding: realistic 100-event dataset for viva. Final documentation and code cleanup.	Both	LOW

5.1 Work Division Between Team Members

Cybersecurity Student	AI/ML Student
AbuseIPDB & GreyNoise API integrations IP reputation scoring pipeline Event validation and deduplication logic CERT-PK advisory parser Crowdsourced report verification pipeline User authentication & role-based access control Admin moderation panel backend SignalR hub configuration and security	Isolation Forest campaign detection algorithm Sector Risk Score ML model Time-series trend analysis and regression Anomaly alert system logic Historical analytics chart generation Chart.js Blazor integration ML microservice (FastAPI) development and deployment Feature engineering for threat classification

6. Conclusion

ThreatCast PK addresses a genuine, documented, and urgent gap in Pakistan's cybersecurity infrastructure. With over 500,000 cyberattacks recorded annually and a 35% surge in cybercrime in 2025, Pakistan's organizations and citizens are operating without any real-time threat intelligence platform. ThreatCast PK is the first publicly accessible, live crowdsourced cyberattack intelligence map purpose-built for Pakistan.

From a course perspective, ThreatCast PK showcases Blazor Server at its best: SignalR enables the real-time map updates that make the platform genuinely live, JavaScript interop powers the Leaflet.js map and Chart.js analytics, and the multi-role CRUD system demonstrates comprehensive data management. The seven implemented CRUD operations span four entities (Attack Reports, User Accounts, Alert Subscriptions, and admin-managed data), each with a clear user-facing purpose directly tied to the platform's core mission.

The technical stack — Blazor Server, SignalR, ASP.NET Core, PostgreSQL, Python FastAPI, AbuseIPDB API, and GreyNoise API — represents an industry-relevant combination of technologies that maps directly to real-world SOC and DFIR tooling. The complete deployment cost is PKR 0, using free tiers from Render.com, Railway, and Supabase, making the project fully reproducible and demonstrable without any financial barrier.

ThreatCast PK is not just a course submission — it is a platform that, once built, provides tangible value to Pakistan's cybersecurity community. Security researchers, journalists, students, and defenders across the country gain their first real-time window into the national threat landscape. We are committed to building it to the highest standard, and look forward to demonstrating it on demo day.

End of Proposal — ThreatCast PK | CS-284L Spring 2026 | Air University Islamabad Campus